
Executable Evaluation Contracts for Target-Weight Trading Pipelines

Kevin Lee 

University of California, Los Angeles
micro1
kevinlee69720@g.ucla.edu

Abstract

A sequence of target weights is not a complete backtest specification. Reported performance also depends on when signals become executable, what residual cash earns, how turnover incurs cost, and whether the comparator bears the same exposure. We encode these choices as executable contracts in QUANTCORTEX, with a canonical target tape, fail-closed invariants, explicit cash and cost accounting, and content-hashed result artifacts.

We first use a retrospective six-ETF rotation case to show how an exact daily identity localizes a negative result: net CAGR is 1.40%, conventional cash-excess Sharpe is -0.15, and active risky allocation contributes -3.38% per year. We then run a protocol frozen in the public repository before the two input panels were retrieved. The expansion evaluates time-series momentum, cross-sectional momentum, short-term reversal, and a walk-forward gradient-boosted model over 2,011 sessions in U.S. sector and country-equity ETF panels. Across the resulting 8 family-panel cells, removing 13-basis-point costs raises annualized arithmetic return by 0.57–1.81 percentage points; all 8 pointwise 95% block-bootstrap intervals lie above zero. Setting residual-cash return to zero produces intervals below zero in 5 cells. Deliberately assigning close-derived targets to the return ending at that close produces intervals below zero in 3 cells and changes 4 within-panel family ranks. Engine conventions create a maximum daily return difference of 93.21 basis points.

These are controlled evaluation effects, not evidence of profitable alpha. The historical panels are neither temporally out of sample nor openly redistributable, and the intervals are pointwise rather than multiplicity-adjusted. The result is a reproducible evaluation methodology and reference implementation with an explicit boundary around its empirical claims.

1 Introduction

A target-weight vector is a convenient interface between prediction, portfolio construction, and execution. Qlib and FinRL use modular research pipelines [Yang et al., 2020, Liu et al., 2020]; FinRL-X extends the interface through broker execution [Yang et al., 2026]. A target, however, is an intention rather than a realized return. A complete evaluation must also state when that target can be held, what uninvested capital earns, how turnover becomes cost, which holdings are credited with each return interval, and what happens when an order outcome is uncertain.

These choices correspond to separate hypotheses. The information set may be causal while the cash account is incomplete. The accounting may be correct while the code is nondeterministic. A reproducible backtest may still offer no investment value, and a valid simulation says little about

whether retrying an uncertain broker request is safe. Treating these questions as one notion of “correctness” obscures the source of failure.

The distinction is particularly important in financial machine learning, where observations are dependent, effective samples are limited, and repeated strategy search creates substantial false-discovery risk [White, 2000, Harvey et al., 2016, Bailey and López de Prado, 2014, Bailey et al., 2017, Arnott et al., 2019]. We ask a narrower systems question: how can a target-weight pipeline make its economic and operational semantics executable, testable, and traceable from the information set to paper-order state?

The primary contribution is the evaluation methodology; the strategies are test vehicles and the software is a reference implementation. Specifically, we (i) define a machine-readable evaluation contract and canonical target tape for information timing, weights, execution lag, cash, costs, comparison, and uncertain order outcomes; (ii) formalize an exact daily return decomposition for changing risky exposure and preserve it under joint block resampling; (iii) estimate five one-switch effects under a repository-frozen protocol across four heterogeneous strategy families and two real-data panels; and (iv) bind every reported result to its protocol, clean source revision, environment, input digest, and artifact hash. The purpose is to expose sensitivity to evaluation semantics, not to select the best-performing strategy after inspection.

2 Related work

Financial backtest validity. Sharpe-ratio inference depends on distributional and serial-dependence assumptions [Lo, 2002]. Data snooping, repeated trials, and researcher degrees of freedom can make a selected backtest appear stronger than the process that generated it [White, 2000, Bailey and López de Prado, 2014, Bailey et al., 2017, Harvey et al., 2016]. Arnott et al. [2019] organize these risks into a research protocol. In a recent preprint, Zhang et al. [2026] estimate decision-time leakage by changing one evaluation convention at a time across multiple models and panels. Our same-close diagnostic is narrower: it is one contract test inside a broader accounting and execution audit, not a general leakage benchmark.

Attribution and matched comparison. Performance evaluation has long separated investment decisions into interpretable return components [Fama, 1972, Brinson and Fachler, 1985]. We apply that principle to a portfolio whose risky exposure changes over time. The identity separates passive exposure, exposure timing around the sample mean, active allocation, and modeled cost. It is an exact arithmetic decomposition, not a new geometric linking method. The strategy set draws on cross-sectional and time-series momentum, short-term reversal, residual momentum, and gradient boosting [Jegadeesh and Titman, 1993, Moskowitz et al., 2012, Lehmann, 1990, Blitz et al., 2011, Friedman, 2001], but each model is used only as a fixed test vehicle.

Research software and implementation risk. Qlib and FinRL provide modular financial-AI workflows [Yang et al., 2020, Liu et al., 2020]; FinRL-X emphasizes a shared weight interface across research and deployment [Yang et al., 2026]. Computational reproducibility remains difficult even with code and data. Across 1,000 computational reproduction attempts in finance, the original code regenerated the reported result exactly only 52% of the time [Pérignon et al., 2024]. In a recent preprint, Yin et al. [2026] compare five engines across 15 strategies and find material divergence from implementation choices, including transaction costs. Our two-engine diagnostic is deliberately more limited: it compares two documented holding conventions on identical target tapes, not independent third-party implementations or an estimate of industry-wide implementation risk. The manifest and source fingerprints follow the broader reproducibility objectives of Pineau et al. [2021]. Our contribution lies in connecting those artifacts to explicit economic contracts within one executable evaluation. No individual invariant or accounting identity is claimed as new; the contribution is their executable integration and controlled stress test.

3 Executable audit protocol

We use “contract” in the design-by-contract sense: an executable invariant with an explicit failure response [Meyer, 1992]. Figure 1 summarizes the evaluation path. Each stage can fail independently, and a later stage does not repair an invalid earlier one. Table 1 states the checks used in this study.

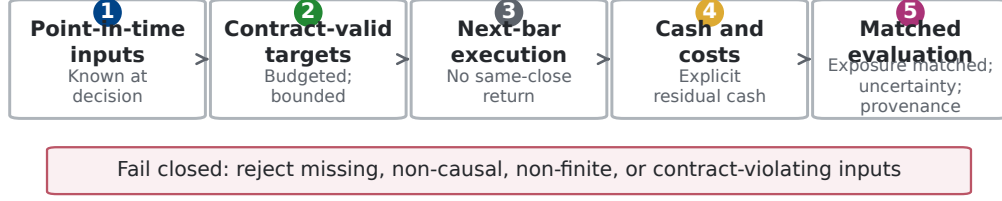


Figure 1: Audit flow from point-in-time inputs to matched evaluation. The audited path validates required inputs, timing, finiteness, and contracts at the stage where each condition becomes testable.

Table 1: Core invariants and failure behavior.

Invariant	Operational check	Failure behavior
Information	Features and fundamentals must be observable by the decision timestamp.	Reject future-dated merges; do not backfill.
Allocation	Allocator output is finite, bounded, and budget-consistent.	Raise a contract error.
Overlay	Timing and risk overlays may reduce exposure but cannot create an invalid book.	Reject non-finite or over-limit weights.
Timing	A close-derived target earns returns only after the first strictly later execution bar.	Shift holdings; never use the observed close twice.
Accounting	Risky assets, residual cash, turnover, and costs share one capital clock.	Reject missing cash bars or cost models.
Comparison	Exact attribution uses realized exposure; a tradable control follows target exposure and pays costs.	Label ex-post controls; charge costs to implementable comparators.
Execution	Persist intent before broker submission; uncertain outcomes are not retried automatically.	Block until broker reconciliation.

3.1 Portfolio, execution, and cash contracts

Let $w_d \in \mathbb{R}^N$ be the risky-asset target decided after observing the close at decision time d , and let $e(d)$ be the first available execution bar strictly after d . At $e(d)$, the primary event engine solves adjusted-close pseudo-shares $q_{i,e(d)}$ against post-cost NAV so that the resulting risky allocation matches w_d . It holds q_i fixed between rebalances. If $P_{i,t}$ is the adjusted close and V_t is portfolio NAV, the realized risky weight at the prior close is $h_{i,t-1} = q_{i,t-1}P_{i,t-1}/V_{t-1}$. Close-derived targets are therefore not filled at the close they observe and do not earn the return ending at their execution close. For the long-only case, $h_{i,t} \geq 0$ and $\sum_i h_{i,t} \leq 1$; the residual cash holding is

$$h_t^c = 1 - \sum_{i=1}^N h_{i,t}. \quad (1)$$

For simple asset returns $r_{i,t}$ over $(t-1, t]$ and cash-account return r_t^c , pre-cost portfolio return is therefore

$$r_t^{\text{gross}} = \sum_{i=1}^N h_{i,t-1} r_{i,t} + h_{t-1}^c r_t^c. \quad (2)$$

Missing cash observations fail the run instead of silently becoming zero. For executed risky-notional change $\Delta a_{i,t}$ measured against pre-trade NAV, define $B_t = \sum_i \max(\Delta a_{i,t}, 0)$ and $S_t = \sum_i \max(-\Delta a_{i,t}, 0)$. One-way turnover is $\max(B_t, S_t)$, gross traded notional is $B_t + S_t$, and the charge as a fraction of pre-trade NAV is $\kappa_t = 0.0013(B_t + S_t)$. If V_t^- is pre-trade NAV and V_{t-1} is prior-close NAV, the return drag is $d_t = \kappa_t V_t^- / V_{t-1}$ and $r_t^{\text{net}} = r_t^{\text{gross}} - d_t$. Target shares are solved against post-cost NAV. The cost model does not separately estimate commissions, bid-ask spread, taxes, or market impact. This proportional model is a first-order sensitivity assumption;

large or urgent orders require models that account for order size, volatility, and execution urgency, such as Almgren and Chriss [2001].

We report nominal CAGR, maximum drawdown, annualized one-way turnover, annualized gross traded notional, and the conventional sample cash-excess Sharpe statistic

$$SR_c = \sqrt{252} \frac{\text{mean}(r_t - r_t^c)}{\text{sd}(r_t - r_t^c)}. \quad (3)$$

Equation 3 is a descriptive statistic, not an iid-normal inference procedure; serial dependence is represented approximately in a separate circular-block sensitivity analysis [Lo, 2002]. An SHV adjusted-close series supplies the return credited to residual cash; the engine does not trade SHV when cash weight changes. It is a short-duration cash proxy, not a frictionless risk-free asset. Because the input contains adjusted closes, the pseudo-shares are total-return accounting units with implicit distribution and corporate-action adjustment, not broker-faithful nominal shares.

3.2 Exact return attribution

Headline performance does not identify which decision created or destroyed value. Let $x_t = \sum_i h_{i,t-1}$ be risky exposure during return period t , let b_t be the return of an equal-initial-weight buy-and-hold basket, and let $\bar{x}$ be the full-sample mean of x_t . Define the daily exposure-matched passive return and the constant-exposure diagnostic as

$$m_t = x_t b_t + (1 - x_t) r_t^c, \quad (4)$$

$$k_t = \bar{x} b_t + (1 - \bar{x}) r_t^c. \quad (5)$$

Then net excess over cash has the exact arithmetic decomposition

$$\begin{aligned} r_t^{\text{net}} - r_t^c = & \underbrace{(r_t^{\text{gross}} - m_t)}_{\text{active risky allocation}} + \underbrace{(m_t - k_t)}_{\text{dynamic exposure timing}} \\ & + \underbrace{(k_t - r_t^c)}_{\text{passive risky exposure}} + \underbrace{(r_t^{\text{net}} - r_t^{\text{gross}})}_{\text{implementation cost}}. \end{aligned} \quad (6)$$

The first term includes group selection, within-risky-asset weighting, and any return effect of the engine’s between-rebalance weight semantics; it is not pure security selection in the Brinson sense. The timing term measures daily exposure changes relative to the sample-average exposure. Because $\bar{x}$ is estimated over the full evaluation window, k_t is an ex-post diagnostic rather than a tradable ex-ante benchmark. Equation 6 holds on every date, so arithmetic annualized means add exactly. CAGR does not admit the same additive interpretation.

We resample all five series in Equation 6 with common circular-block indices. Consequently, the identity also holds within every bootstrap draw. The reported fraction of positive draws is a resampling diagnostic, not a posterior probability or a multiple-testing-adjusted p -value.

3.3 Single-assumption diagnostics

Three diagnostics change one assumption while holding the targets, primary engine, and input matrix fixed: residual cash earns zero, proportional costs are removed, or a close-derived target is deliberately executed one bar before that close so it earns the return ending at the observed close. The last construction uses future information and is therefore invalid. Under explicit share holdings, changing the cash return can also change realized exposure between rebalances through the NAV denominator. These are sensitivity checks, not candidate strategies, and they play no role in model selection. The paired design resembles the one-switch logic of Zhang et al. [2026], but here the switches audit cash, costs, and execution semantics in one pipeline.

3.4 Reference implementation and order-state controls

The reference implementation separates data, alpha, portfolio, timing, risk, backtest, execution, and strategy layers. Allocator outputs satisfy a strict budget contract. Post-overlay outputs use a relaxed exposure contract because a timing or risk gate may intentionally leave capital in cash. Point-in-time

merge guards reject fundamentals that were not yet public, and optional broker, machine-learning, and provider SDKs remain lazy imports so the scientific core can be tested offline.

A central manifest records the clean source revision, Python and package versions, dependency lock, input SHA-256, experiment design, source-tree hashes, and the hash of every published table and figure. A versioned JSON contract and canonical long-form target-tape schema make the engine boundary machine readable. Each empirical run round-trips every strategy decision stream through the versioned payload boundary and records a canonical payload hash; deterministic fixtures test the same boundary without using the empirical price matrix. The empirical sections below exercise the research and backtest contracts. Order-state controls are verified by deterministic tests: paper-trading state is written atomically as a versioned snapshot, intent is persisted before submission, and an uncertain submission is not retried automatically. SDK-conformance tests instantiate the pinned Alpaca and Interactive Brokers request types. They do not test authenticated transport or venue behavior. Appendix G separates exact identities, property tests, counterfactuals, fault injection, SDK conformance, and untested live behavior.

4 Evaluation design

4.1 Retrospective worked audit

The first study is a fixed six-ETF weekly rotation strategy evaluated from 2018-01-02 through 2025-12-31. It combines group ranking, residual momentum, a Gaussian-mixture exposure gate, and volatility scaling. The study was assembled before a prospective protocol existed, and its earlier trial count is unknown; it is therefore a worked accounting audit, not confirmatory evidence. Its known construct issues are retained rather than repaired after seeing the result: QQQ is both a group member and the ranking benchmark, two regime features are exact volatility rescalings, early regime windows are immature, and one seed controls the mixture fit. Appendix A gives the complete configuration.

The owner-supplied adjusted-close matrix is not distributed. It is identified by a manifest-recorded SHA-256. The event engine credits SHV to residual cash, charges 13 basis points per dollar traded on each buy or sell leg, and executes close-derived weekly targets on the first strictly later bar. The exact attribution uses a realized-exposure ex-post control; a separate causal target-exposure comparator pays the same modeled costs.

4.2 Repository-frozen expansion

The second study was specified in the public repository before its two provider requests. The freeze is a Git timestamp, not an external registry entry, and the historical interval is not temporally out of sample. Two complete-row panels use adjusted closes from 2014-01-02 through 2025-12-31: nine U.S. sector ETFs and ten country-equity ETFs, each with SHV. Evaluation covers 2,011 sessions from 2018-01-02 through 2025-12-31. There is no forward fill or symbol substitution. The panels contain surviving, currently available ETFs and share broad market exposures, so they are not independent replications.

Table 2 summarizes the frozen strategy families. Each decision uses the first complete session of the month and executes on the first strictly later panel close. Long-only target exposure is at most one; unused capital earns SHV. The primary event engine holds adjusted-close pseudo-shares between rebalances and charges 13 basis points per dollar traded. Every target round-trips through the canonical JSON tape before evaluation.

The learned family pools asset-month observations without symbol identity. Features are 5-, 21-, 63-, 126-, and 252-session log returns; 21- and 63-session sample volatility; distance from the trailing 252-session high; and ordinal cross-sectional ranks of 21- and 252-session returns. They are neither standardized nor winsorized. At each evaluation decision the model uses only training labels whose 21-session endpoint is observable by that close, retains at most 60 mature decision months, and requires at least 24. A gradient-boosted regressor with 100 depth-2 trees, learning rate 0.03, minimum leaf size 10, and subsample 0.8 is refit separately for seeds 11, 29, 47, 71, and 97. Features and labels are recomputed walk-forward; no fitted model crosses a decision boundary.

Table 2: Frozen expansion strategies. Ties use ascending symbols.

Family	Fixed rule
Time-series momentum	Positive 252-session total returns; equal weight across eligible assets.
Cross-sectional momentum	Top three returns from $t - 252$ to $t - 21$; one-third each.
Short-term reversal	Up to three lowest negative 5-session returns; one-third each, without renormalizing unused exposure.
Learned GBRT	Walk-forward 21-session return prediction from return, volatility, trailing-high, and rank features; top three positive predictions; five frozen seeds.

4.3 One-switch estimands and uncertainty

For each family–panel cell, five paired switches compare (i) invalid same-close assignment with next-bar execution, (ii) zero cash return with SHV, (iii) zero cost with 13-basis-point cost, (iv) the strategy with a causal costed equal-weight comparator at the same target risky exposure, and (v) vectorized with event-driven accounting. The comparator is conditional on the strategy’s target exposure and is a conditional exposure control, not an independent market benchmark. The vectorized engine re-pegs weights between decisions, whereas the event engine holds drifting pseudo-shares; their difference is a model-convention sensitivity, not automatically an implementation error.

Primary outcomes are paired changes in annualized arithmetic return and conventional SHV-excess Sharpe. We use 5,000 joint circular-block draws with a 21-session primary block and repeat the calculation with 5 and 63 sessions [Politis and Romano, 1992]. Return series, cash, and all five learned seeds share each draw. Intervals are unstudentized pointwise 95% percentile intervals. They condition on the selected panels and frozen models, are not multiplicity adjusted, and do not establish future performance.

5 Results

5.1 Baseline outcomes are inputs to the audit, not discoveries

Appendix Figure 5 reports after-cost arithmetic return and SHV-excess Sharpe. 7 of 8 family–panel cells have positive sample Sharpe; country-panel time-series momentum is slightly negative. The sector panel is stronger for every family. These rankings are descriptive: the panels, families, and 2018–2025 interval are not a random sample, and no baseline interval or multiple-testing correction supports an alpha claim.

5.2 Contract effects depend on the strategy

Table 3 counts pointwise intervals across the eight family–panel cells. Cost removal is a positive-by-construction check, but its size is material: annualized arithmetic return rises by 0.57–1.81 percentage points. In contrast, same-close assignment is not a uniformly optimistic bias. Its interval is below zero in 3 cells and overlaps zero in 5, driven by the deliberately contrarian reversal rule and, in one panel, the learned model. Zero cash return is below zero in 5 cells and overlaps zero in 3. These patterns show why a convention cannot be summarized by one global correction factor.

Table 3: Number of 21-session pointwise intervals below zero / overlapping zero / above zero, out of eight family–panel cells. Counts are descriptive and are not multiplicity-adjusted tests.

Switch difference	Annual return	Sharpe
Same-close - next-bar	3/5/0	3/5/0
Zero cash - SHV cash	5/3/0	5/3/0
Zero cost - 13 bp cost	0/0/8	0/0/8
Strategy - costed comparator	1/7/0	1/7/0
Vectorized - event-driven	0/4/4	0/4/4

Figure 2 displays the return effects. Relative to its costed exposure-matched comparator, only country-panel time-series momentum has an interval below zero; the other seven overlap zero. This

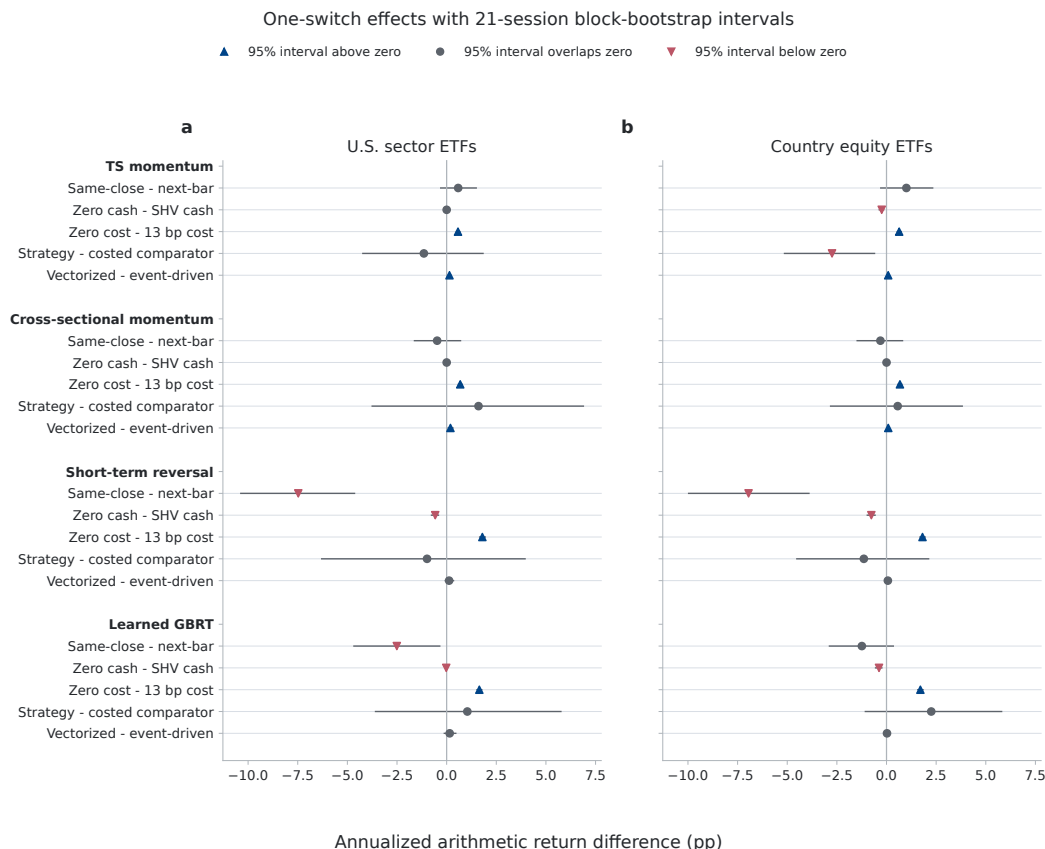


Figure 2: Paired annualized arithmetic-return effects with 21-session circular-block percentile intervals. Marker shape and color encode whether the pointwise interval is above, overlaps, or is below zero. The same-close switch is causally invalid and is never a candidate strategy.

does not validate the strategies. It says the conditional comparator is not precise enough to rank most cells on this single path. Same-close and zero-cost switches change 4 and 4 of the 8 within-panel family ranks, respectively; replacing strategies with their costed comparators changes 5 ranks.

5.3 Seed and engine sensitivity remain visible

The learned family is not represented by a favorable seed. Across the five frozen seeds, baseline SHV-excess Sharpe ranges from 0.47–0.60 in sectors and 0.27–0.37 in countries; every seed-level result is published. The vectorized-minus-event mean effect is small, but the maximum one-day difference reaches 93.21 basis points and the largest absolute terminal wealth gap is 3.89%. This divergence is expected from daily weight re-pegging versus drifting pseudo-shares. Appendix Figures 7 and 8 report both diagnostics.

5.4 The worked audit localizes one negative result

The retrospective case records 1.40% net CAGR and -0.15 SHV-excess Sharpe. Its causal target-exposure comparator, after its own costs, records 5.72% and 0.62; annualized arithmetic shortfall is -4.14% [-6.70%, -1.63%]. Equation 6 attributes net excess over SHV exactly: passive risky exposure contributes 3.90%, active risky allocation -3.38%, dynamic exposure timing 0.30%, and modeled cost -1.72% per year. The active-allocation interval [-5.85%, -0.90%] is wholly below zero.

Crediting SHV instead of zero to residual cash raises CAGR from -0.17% to 1.40% because mean risky exposure is only 30.14%. Omitting modeled costs changes Sharpe from -0.15 to 0.14; invalid same-close assignment changes it to 0.03. The expansion qualifies that last result: look-ahead

changes estimates and ranks, but its direction depends on the signal. Appendix A.1 contains accounting, cost, subperiod, and block-length diagnostics for the worked case.

6 Limitations and broader impact

6.1 Statistical validity

The retrospective strategy predates the protocol, and no complete log of its earlier trials survives. The expansion protocol was committed before the two provider requests, but it was not externally registered and evaluates an historical interval. Neither study is a temporal holdout. A Deflated Sharpe Ratio or Reality Check would require a defensible count of prior trials that is not available [White, 2000, Bailey and López de Prado, 2014].

The eight family-panel cells are dependent: models share dates, panels share global market exposures, and five learned seeds are repeated fits rather than independent strategies. The circular-block bootstrap conditions on these paths and approximates serial dependence through fixed block lengths. Pointwise intervals and interval-category counts are not multiplicity adjusted. Stability across 5, 21, and 63 sessions is sensitivity evidence, not proof of stationarity or future performance. The conventional $\sqrt{252}$ Sharpe remains a descriptive statistic; block resampling does not remove the assumptions needed to interpret it. Component interval endpoints need not add, although the exact decomposition holds within every joint draw.

6.2 Economic construct validity

“Active risky allocation” in the worked case combines group selection, within-risky weighting, and between-rebalance engine semantics. “Dynamic exposure timing” is relative to full-sample mean exposure, hence diagnostic rather than tradable. In both studies, the exposure-matched comparator is conditional on the strategy’s own target path; it controls exposure but is not an independent benchmark. Arithmetic means decompose exactly; compounded wealth does not.

The frozen models are deliberately simple. The ETF panels do not represent point-in-time constituent universes, the country funds are unhedged U.S.-listed securities, and the learned model pools symbols without identity or hyperparameter search. SHV has duration, expenses, and market-price variation; it is not a risk-free account. Adjusted-close pseudo-shares embed total-return adjustments and omit broker share records, spreads, queue position, and fill uncertainty. The proportional cost model has no order-size, volatility, venue, impact, ADV, or capacity term.

6.3 External validity and reproducibility

The expansion broadens the worked case but still covers two U.S.-listed ETF panels, four hand-specified families, one eight-year interval, and two internal engines. It does not estimate a population distribution of backtest errors or match broader multi-model and multi-engine studies [Zhang et al., 2026, Yin et al., 2026]. Provider-adjusted histories can be revised. Raw inputs are not redistributed, and provider permission for the derived publication has not been independently verified. Open schemas, canonical tapes, and conformance fixtures reproduce software semantics, not the reported returns; full empirical reproduction requires owner-supplied matrices with the recorded hashes.

6.4 Operational scope and broader impact

Fault-injection tests cover stale or uncertain execution state, but no live authentication, venue behavior, funded order, capacity, or production control was tested. The intended benefit is reduced false confidence; the principal risk is mistaking an audited simulation for investment advice or deployment approval. Paper-mode defaults and fail-closed retries mitigate but do not remove that risk. Past simulated performance does not imply future returns.

7 Conclusion

A target-weight generator does not determine a realized return; information timing, execution, cash, costs, and comparator exposure complete the experiment. In the retrospective case, exact attribution

shows that active allocation and modeled cost erase the benefit of passive risky exposure. In the frozen expansion, removing modeled cost raises annualized return in every family–panel cell, while cash, same-close timing, comparator, and engine effects vary by strategy. The same-close switch can worsen a contrarian model rather than mechanically inflate all results; the error is using unavailable information, not a guaranteed sign of bias. Canonical tapes, one-switch counterfactuals, and artifact provenance make those distinctions testable. The contribution is an executable evaluation methodology and bounded negative evidence, not a profitable signal or deployment claim.

References

- Robert Almgren and Neil Chriss. Optimal execution of portfolio transactions. *The Journal of Risk*, 3(2):5–39, 2001. doi: 10.21314/JOR.2001.041.
- Rob Arnott, Campbell R. Harvey, and Harry Markowitz. A backtesting protocol in the era of machine learning. *The Journal of Financial Data Science*, 1(1):64–74, 2019. doi: 10.3905/jfds.2019.1.064.
- David H. Bailey and Marcos López de Prado. The deflated Sharpe ratio: Correcting for selection bias, backtest overfitting, and non-normality. *The Journal of Portfolio Management*, 40(5):94–107, 2014. doi: 10.3905/jpm.2014.40.5.094.
- David H. Bailey, Jonathan M. Borwein, Marcos López de Prado, and Qiji Jim Zhu. The probability of backtest overfitting. *The Journal of Computational Finance*, 20(4):39–69, 2017. doi: 10.21314/JCF.2016.322.
- David Blitz, Joop Huij, and Martin Martens. Residual momentum. *Journal of Empirical Finance*, 18(3):506–521, 2011. doi: 10.1016/j.jempfin.2011.01.003.
- Gary P. Brinson and Nimrod Fachler. Measuring non-U.S. equity portfolio performance. *The Journal of Portfolio Management*, 11(3):73–76, 1985. doi: 10.3905/jpm.1985.409005.
- Eugene F. Fama. Components of investment performance. *The Journal of Finance*, 27(3):551–567, 1972. doi: 10.1111/j.1540-6261.1972.tb00984.x.
- Jerome H. Friedman. Greedy function approximation: A gradient boosting machine. *The Annals of Statistics*, 29(5):1189–1232, 2001. doi: 10.1214/aos/1013203451.
- Campbell R. Harvey, Yan Liu, and Heqing Zhu. ... and the cross-section of expected returns. *The Review of Financial Studies*, 29(1):5–68, 2016. doi: 10.1093/rfs/hhv059.
- Narasimhan Jegadeesh and Sheridan Titman. Returns to buying winners and selling losers: Implications for stock market efficiency. *The Journal of Finance*, 48(1):65–91, 1993. doi: 10.1111/j.1540-6261.1993.tb04702.x.
- Bruce N. Lehmann. Fads, martingales, and market efficiency. *The Quarterly Journal of Economics*, 105(1):1–28, 1990. doi: 10.2307/2937816.
- Xiao-Yang Liu, Hongyang Yang, Qian Chen, Runjia Zhang, Liuqing Yang, Bowen Xiao, and Christina Dan Wang. FinRL: A deep reinforcement learning library for automated stock trading in quantitative finance. *arXiv preprint arXiv:2011.09607*, 2020. URL <https://arxiv.org/abs/2011.09607>.
- Andrew W. Lo. The statistics of Sharpe ratios. *Financial Analysts Journal*, 58(4):36–52, 2002. doi: 10.2469/faj.v58.n4.2453.
- Bertrand Meyer. Applying “design by contract”. *Computer*, 25(10):40–51, 1992. doi: 10.1109/2.161279.
- Tobias J. Moskowitz, Yao Hua Ooi, and Lasse Heje Pedersen. Time series momentum. *Journal of Financial Economics*, 104(2):228–250, 2012. doi: 10.1016/j.jfineco.2011.11.003.

- Christophe Pérignon, Olivier Akmansoy, Christophe Hurlin, Anna Dreber, Felix Holzmeister, Jürgen Huber, Magnus Johannesson, Michael Kirchler, Albert J. Menkveld, Michael Razen, and Utz Weitzel. Computational reproducibility in finance: Evidence from 1,000 tests. *The Review of Financial Studies*, 37(11):3558–3593, 2024. doi: 10.1093/rfs/hhae029.
- Joelle Pineau, Philippe Vincent-Lamarre, Koustuv Sinha, Vincent Larivière, Alina Beygelzimer, Florence d’Alché-Buc, Emily Fox, and Hugo Larochelle. Improving reproducibility in machine learning research: A report from the NeurIPS 2019 reproducibility program. *Journal of Machine Learning Research*, 22(164):1–20, 2021. URL <https://jmlr.org/papers/v22/20-303.html>.
- Dimitris N. Politis and Joseph P. Romano. A circular block-resampling procedure for stationary data. In Raoul LePage and Lynne Billard, editors, *Exploring the Limits of Bootstrap*, pages 263–270. John Wiley & Sons, New York, 1992.
- Halbert White. A reality check for data snooping. *Econometrica*, 68(5):1097–1126, 2000. doi: 10.1111/1468-0262.00152.
- Hongyang Yang, Boyu Zhang, Yang She, Xinyu Liao, and Xiaoli Zhang. FinRL-X: An AI-native modular infrastructure for quantitative trading. *arXiv preprint arXiv:2603.21330*, 2026. URL <https://arxiv.org/abs/2603.21330>.
- Xiao Yang, Weiqing Liu, Dong Zhou, Jiang Bian, and Tie-Yan Liu. Qlib: An AI-oriented quantitative investment platform. *arXiv preprint arXiv:2009.11189*, 2020. URL <https://arxiv.org/abs/2009.11189>.
- Dong Yin, Takeshi Miki, Vladislav Lesnichenko, and Vasyl Gural. Implementation risk in portfolio backtesting: A previously unquantified source of error. *arXiv preprint arXiv:2603.20319*, 2026. URL <https://arxiv.org/abs/2603.20319>.
- Fan Zhang, Zhen Li, Sijia Peng, and Yu Chen. When alpha disappears: A one-switch benchmark for decision-time leakage in financial backtests. *arXiv preprint arXiv:2605.23959*, 2026. URL <https://arxiv.org/abs/2605.23959>.

A Retrospective case specification

The universe is QQQ, VGT, GLD, TLT, SPY, and VIG, grouped as growth equities, gold and nominal duration, and broad and dividend equities. On the first session of each week, the strategy keeps the top two groups by a 126-session information ratio relative to QQQ. Within selected groups, it estimates a one-factor regression over 126 sessions and ranks a separate 126-session residual-return window with a 21-session gap. Positive scores receive proportional long-only weights capped at 60% per asset.

A seeded three-component Gaussian mixture maps QQQ return and two exact rescalings of realized volatility to zero, half, or full exposure. A separate 21-session volatility scaler targets 20 volatility points. Named variants remove the regime gate, the volatility scaler, or both; none was selected after the result. The full manifest records every parameter and warm-up requirement.

A.1 Retrospective accounting results

Table 4: Audited retrospective accounting over 2018–2025.

Series	CAGR	Sharpe	Max drawdown
Strategy, after costs and SHV cash	1.40%	-0.15	-9.83%
Strategy, before costs and SHV cash	3.17%	0.14	-8.60%
SHV cash proxy	2.50%	–	-0.45%
Realized-exposure control, gross	6.74%	0.80	-4.64%
Target-exposure comparator, after costs	5.72%	0.62	-4.94%
Strategy, after costs and zero cash	-0.17%	0.00	-15.15%

Table 5: Exact attribution of annualized arithmetic net excess over SHV in the retrospective case. Intervals use common 21-session block draws.

Component	Mean	95% interval	Positive draws
Active risky allocation	-3.38%	[-5.85%, -0.90%]	15/5000
Dynamic exposure timing	0.30%	[-2.57%, 3.26%]	2889/5000
Passive risky exposure	3.90%	[0.82%, 6.85%]	4963/5000
Modeled implementation cost	-1.72%	[-2.08%, -1.38%]	0/5000
Net excess over cash	-0.90%	[-4.77%, 3.14%]	1682/5000

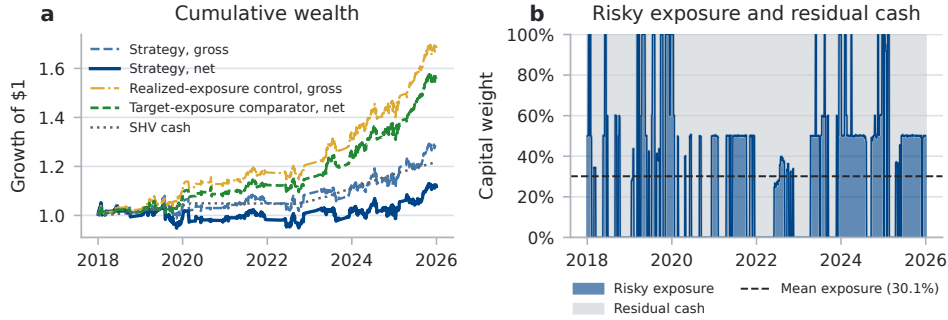


Figure 3: Growth of one dollar and the capital split between risky assets and SHV under the audited retrospective accounting.

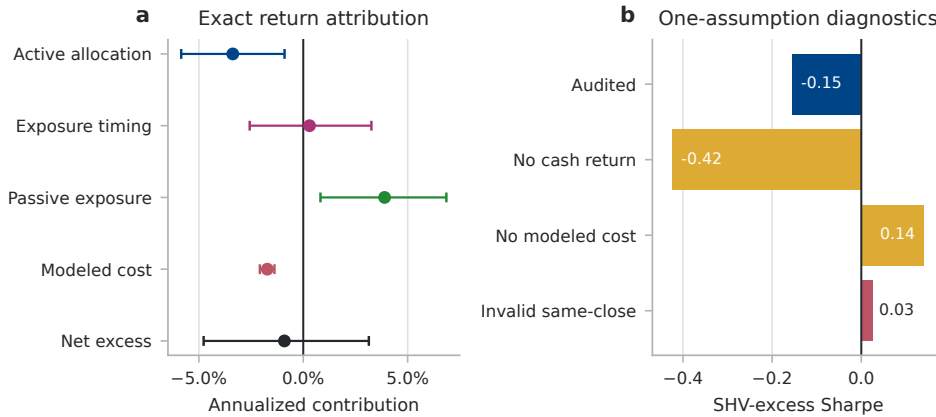


Figure 4: Exact return attribution and one-assumption diagnostics for the retrospective case. The same-close result is causally invalid.

B Reproducibility details

The retrospective and expansion generators are `scripts/run_paper_experiments.py` and `scripts/run_expansion_experiments.py`. Their release wrappers check out the recorded source revision in a detached clean worktree, keep raw inputs outside that worktree, and verify source cleanliness and output hashes. The reviewed invocations are:

```
scripts/release_expansion_artifacts.sh \
  local_data/expansion
scripts/release_paper_artifacts.sh \
  local_data/published_rotation_prices.csv
```

Exact empirical reproduction requires three owner-supplied files matching the recorded SHA-256 values: the six-ETF retrospective matrix and the two expansion panels. Aggregate CSVs, generated LaTeX, figures, and hashes are listed in `paper/results/manifest.json` and `paper/expansion/results/manifest.json`. The expansion manifest also records the pre-data

protocol commit, provider-request metadata, canonical tape hashes, thread libraries, dependency lock, and every source file used by the run. The three matrix digests are:

Retrospective: efb384a62157e56a0cd8065abf45c1ed07d90ec26c681e5d54d74fe4cb9c55e1
U.S. sectors: c5947c4ca6ad6d21ad834c8f344dcdd07acc59ba411e3dcb2202a2413642b2f9
Country equities: 870fa926b5080378c65bd629b9959bd73b16391dded5720c894c0f35558132a9

The expansion protocol digest is

e49e41a12a19fa5404a573ba5e21eb8a2888e616985f8c610d9652866923315c

The frozen JSON key `cost_per_one_way_gross_notional` is misnamed: its 0.0013 value is applied to gross two-sided traded notional, as defined in Section 3. The key is retained to preserve the pre-retrieval protocol hash. The retrospective source-tree digest is:

e5cbb756404c6776eb5945a1590664bb91b22dc7b0daea22efbf80a5be78c6b2

The public code is available at <https://github.com/magnaquant/quantcortex>. Both reviewed runs were CPU-only; environment versions and thread-pool metadata are machine readable. Supported CI runs use Python 3.11 through 3.14. Repository-wide testing, packaging, notebook execution, and broker conformance checks are engineering validation rather than model search.

C Expansion tables and diagnostics

Panel	Family	Ann. return	Sharpe	CAGR	Max DD
U.S. sector ETFs	TS momentum	11.44%	0.49	10.29%	-34.16%
U.S. sector ETFs	Cross-sectional momentum	14.19%	0.60	13.04%	-32.06%
U.S. sector ETFs	Short-term reversal	11.02%	0.44	9.54%	-45.08%
U.S. sector ETFs	Learned GBRT	13.66%	0.53	12.10%	-39.96%
Country equity ETFs	TS momentum	2.18%	-0.02	0.88%	-40.45%
Country equity ETFs	Cross-sectional momentum	8.84%	0.33	7.21%	-40.73%
Country equity ETFs	Short-term reversal	6.03%	0.20	4.48%	-43.91%
Country equity ETFs	Learned GBRT	7.84%	0.31	6.56%	-33.56%

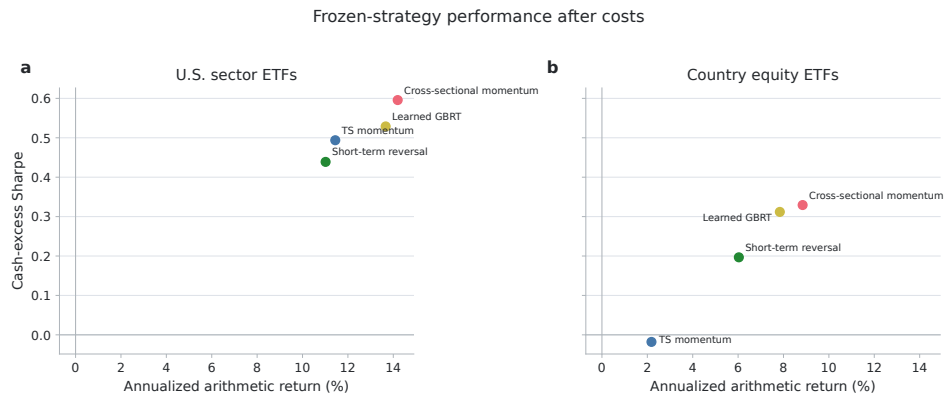


Figure 5: After-cost baseline estimates under the frozen expansion protocol. The learned point is the arithmetic mean across five seed-level metrics.

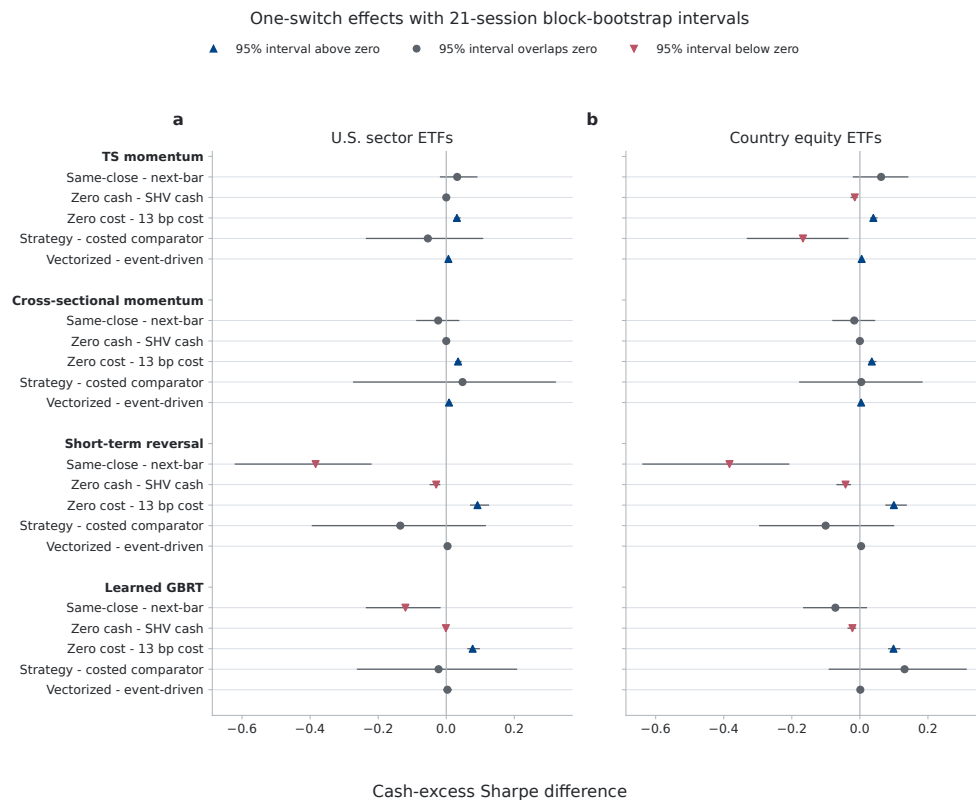


Figure 6: Paired conventional cash-excess Sharpe effects and 21-session pointwise block-bootstrap intervals for all expansion cells.

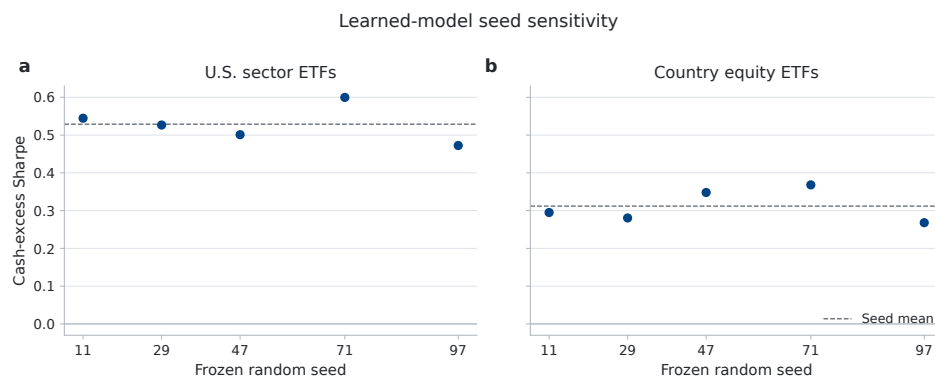


Figure 7: Learned-model baseline Sharpe for all five frozen random seeds. Dashed lines are within-panel seed means.

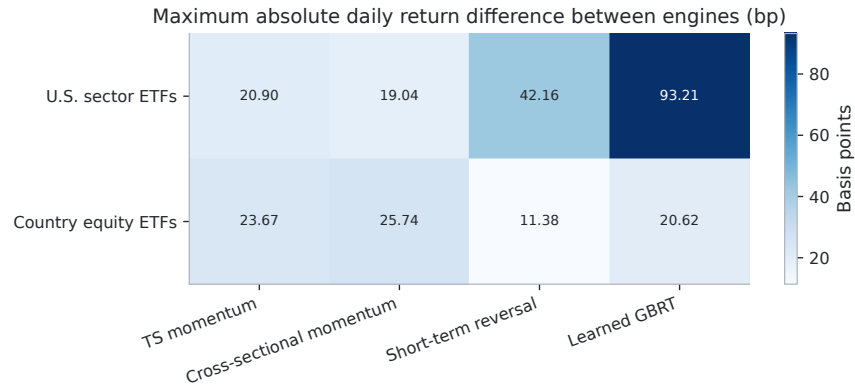


Figure 8: Maximum absolute one-day return difference between event-driven pseudo-share accounting and the vectorized daily re-pegging convention.

D Retrospective block-length sensitivity

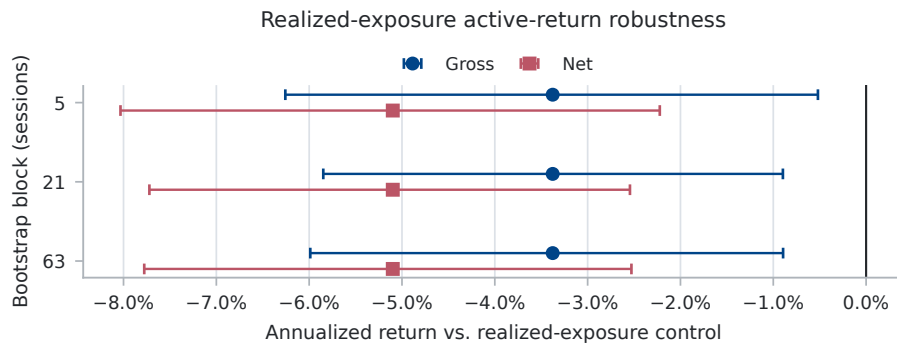


Figure 9: Active-return estimates and 95% percentile intervals under 5-, 21-, and 63-session circular-block resampling against the realized-exposure attribution control. Gross compares before-cost strategy returns with the gross control; net retains modeled strategy costs while the control remains gross.

E Retrospective engine comparison

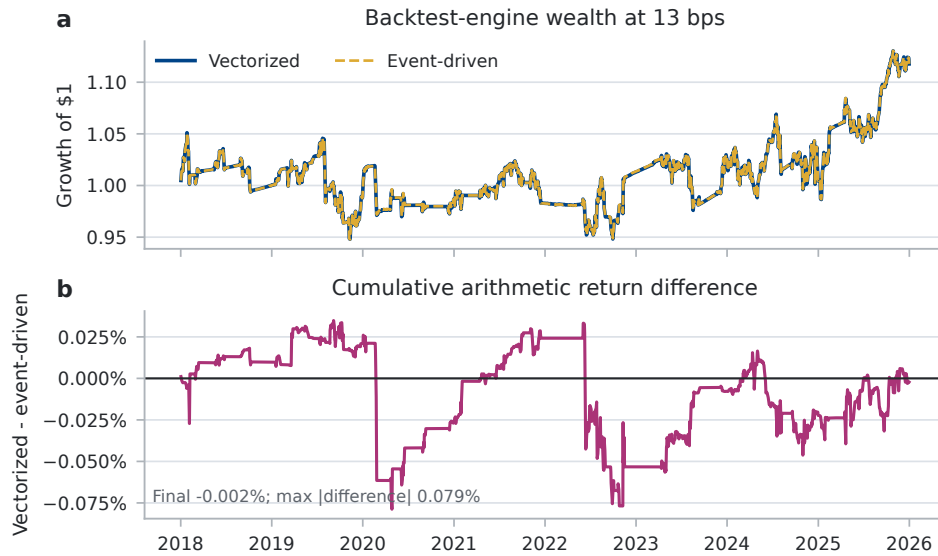


Figure 10: The vectorized engine re-pegs target weights between rebalances, whereas the event-driven engine holds explicit drifting adjusted-close pseudo-shares. Their small difference here does not remove the semantic distinction.

F Retrospective cost and overlay sensitivity

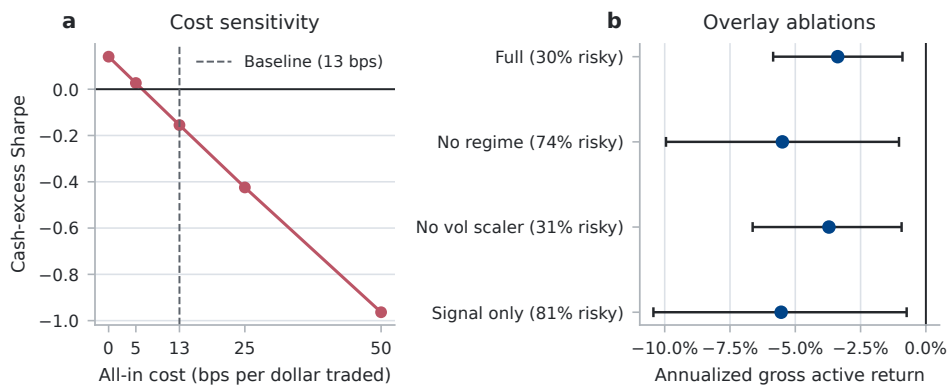


Figure 11: (a) After-cost Sharpe sensitivity for the full strategy. (b) Annualized gross strategy return minus the realized-exposure attribution control, with 95% intervals from common 21-session circular-block draws. Tick labels report mean risky exposure; all named audit variants are shown.

G Contract-to-evidence map

The artifact distinguishes evidence types rather than treating every passing check as equivalent:

Evidence class	Claim supported	Committed evidence
Exact identity	Daily return components reconstruct net cash excess	Per-row assertion and <code>return_decomposition.csv</code>
Property test	Weight, timing, cash, and target-tape invariants hold	Deterministic pytest cases, including hand-computable tapes
Counterfactual	One declared assumption changes the reported metric	<code>protocol_switches.csv</code> and <code>contract_effects.csv</code>
Fault injection	Unsafe or stale execution state fails closed	Truncated reconciliation, unknown status, and stale-writer tests
SDK conformance	Offline request and response mappings match pinned SDKs	Broker model-construction tests and behavioral mocks
Untested live	Authenticated transport, venue state, and funded orders	No supporting evidence; explicitly outside scope

H Use of language-model assistance

The author used LLM-based coding assistants for implementation planning, repository inspection, code drafting, adversarial review, regression tests, and copyediting. The assistants are not an evaluated model or data source. The author approved and publicly froze the expansion protocol before its provider requests, reviewed the code and artifacts, and remains responsible for the methods and claims. Numerical claims are generated by committed scripts and machine-readable artifacts.

NeurIPS Paper Checklist

1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [\[Yes\]](#)

Justification: The abstract and Introduction identify an evaluation methodology, distinguish the retrospective worked audit from the repository-frozen expansion, and avoid alpha, population-benchmark, or deployment claims. Reported values are generated from the artifacts used in Sections 4 and 5.

Guidelines:

- The answer [\[N/A\]](#) means that the abstract and introduction do not include the claims made in the paper.
- The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [\[No\]](#) or [\[N/A\]](#) answer to this question will not be perceived well by the reviewers.
- The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
- It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [\[Yes\]](#)

Justification: Section 6 discusses dependent panels and strategy cells, the historical rather than temporal holdout, unknown retrospective trials, pointwise intervals, comparator and engine semantics, ETF-universe selection, cash and cost proxies, data availability, and untested live transport.

Guidelines:

- The answer [\[N/A\]](#) means that the paper has no limitation while the answer [\[No\]](#) means that the paper has limitations, but those are not discussed in the paper.
- The authors are encouraged to create a separate “Limitations” section in their paper.

- The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
- The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
- The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
- The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
- If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
- While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren't acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

3. Theory assumptions and proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [N/A]

Justification: The paper contains accounting definitions and empirical estimators, but no theorem, lemma, or new theoretical result requiring a proof.

Guidelines:

- The answer [N/A] means that the paper does not include theoretical results.
- All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
- All assumptions should be clearly stated or referenced in the statement of any theorems.
- The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
- Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
- Theorems and Lemmas that the proof relies upon should be properly referenced.

4. Experimental result reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [No]

Justification: Sections 3–4 and the appendices disclose algorithms, dates, symbols, hyperparameters, seeds, versions, release commands, and input hashes for both studies. The three raw price matrices are not redistributed, so independent bitwise reproduction of the empirical returns is not fully open.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.

- If the paper includes experiments, a [No] answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
- If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.
- Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general, releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.
- While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
 - (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
 - (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
 - (c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).
 - (d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. Open access to data and code

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [No]

Justification: The MIT-licensed code, schemas, fixtures, aggregate results, figures, and manifests are open, but the provider-derived adjusted-close matrices are not. The reproducibility appendix gives exact commands and required local inputs.

Guidelines:

- The answer [N/A] means that paper does not include experiments requiring code.
- Please see the NeurIPS code and data submission guidelines (<https://neurips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- While we encourage the release of code and data, we understand that this might not be possible, so [No] is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).
- The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (<https://neurips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
- The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
- At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).

- Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. Experimental setting/details

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

Answer: [\[Yes\]](#)

Justification: Section 4 specifies both studies' panels, dates, decision timing, strategy rules, learned-model features and hyperparameters, five seeds, cash and cost rules, comparators, engines, bootstrap settings, and retrospective trial-count limitation.

Guidelines:

- The answer [\[N/A\]](#) means that the paper does not include experiments.
- The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
- The full details can be provided either with the code, in appendix, or as supplemental material.

7. Experiment statistical significance

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [\[Yes\]](#)

Justification: Sections 3–5 report joint circular-block intervals for paired arithmetic-return effects and recomputed conventional sample Sharpe differences, with 5-, 21-, and 63-session block sensitivity. The paper states the 5,000 replications, shared-draw construction, dependence rationale, and pointwise, non-multiplicity-adjusted interpretation.

Guidelines:

- The answer [\[N/A\]](#) means that the paper does not include experiments.
- The authors should answer [\[Yes\]](#) if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.
- The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
- The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
- The assumptions made should be given (e.g., Normally distributed errors).
- It should be clear whether the error bar is the standard deviation or the standard error of the mean.
- It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
- For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g., negative error rates).
- If error bars are reported in tables or plots, the authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

8. Experiments compute resources

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [\[Yes\]](#)

Justification: The reproducibility appendix identifies CPU-only execution, supported Python versions, machine-readable environment and thread-pool metadata, and distinguishes experiment runs from repository-wide engineering validation. Reviewed wall times and hardware are recorded with the release notes.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
- The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.
- The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

9. Code of ethics

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics <https://neurips.cc/public/EthicsGuidelines>?

Answer: [No]

Justification: The work uses no human subjects or private personal data and places no live orders. However, as disclosed in Sections 4 and 6 and item 12, provider permission for publication of the derived aggregates has not been independently verified, so conformity cannot be certified in every respect.

Guidelines:

- The answer [N/A] means that the authors have not reviewed the NeurIPS Code of Ethics.
- If the authors answer [No], they should explain the special circumstances that require a deviation from the Code of Ethics.
- The authors should make sure to preserve anonymity (e.g., if there is a special consideration due to laws or regulations in their jurisdiction).

10. Broader impacts

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [Yes]

Justification: Section 6 discusses the benefit of reducing false confidence and the risk of treating simulated results as investment advice or deployment approval, together with practical mitigations.

Guidelines:

- The answer [N/A] means that there is no societal impact of the work performed.
- If the authors answer [N/A] or [No], they should explain why their work has no societal impact or why the paper does not address societal impact.
- Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.
- The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate to point out that an improvement in the quality of generative models could be used to generate Deepfakes for disinformation. On the other hand, it is not needed to point out that a generic algorithm for optimizing neural networks could enable people to train models that generate Deepfakes faster.
- The authors should consider possible harms that could arise when the technology is being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following from (intentional or unintentional) misuse of the technology.
- If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

11. Safeguards

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

Answer: [N/A]

Justification: The paper releases no pretrained model, scraped media corpus, personal data, or other high-risk dual-use asset requiring gated access.

Guidelines:

- The answer [N/A] means that the paper poses no such risks.
- Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.
- Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.
- We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.

12. Licenses for existing assets

Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

Answer: [No]

Justification: The manifests identify Yahoo Finance via yfinance, record request settings and provider notices, and confirm that raw quotes are not redistributed. The project has not independently verified provider authorization for public publication of the derived aggregates, so it cannot certify full terms compliance.

Guidelines:

- The answer [N/A] means that the paper does not use existing assets.
- The authors should cite the original paper that produced the code package or dataset.
- The authors should state which version of the asset is used and, if possible, include a URL.
- The name of the license (e.g., CC-BY 4.0) should be included for each asset.
- For scraped data from a particular source (e.g., website), the copyright and terms of service of that source should be provided.
- If assets are released, the license, copyright information, and terms of use in the package should be provided. For popular datasets, paperswithcode.com/datasets has curated licenses for some datasets. Their licensing guide can help determine the license of a dataset.
- For existing datasets that are re-packaged, both the original license and the license of the derived asset (if it has changed) should be provided.
- If this information is not available online, the authors are encouraged to reach out to the asset's creators.

13. New assets

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer: [Yes]

Justification: The repository documents the MIT-licensed code, target-tape and evaluation-contract schemas, conformance fixtures, frozen expansion protocol, architecture, tests, release commands, aggregate tables, figures, provenance manifests, and known limitations.

Guidelines:

- The answer [N/A] means that the paper does not release new assets.

- Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license, limitations, etc.
- The paper should discuss whether and how consent was obtained from people whose asset is used.
- At submission time, remember to anonymize your assets (if applicable). You can either create an anonymized URL or include an anonymized zip file.

14. Crowdsourcing and research with human subjects

Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

Answer: [N/A]

Justification: The research includes no crowdsourcing and no human subjects.

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.
- According to the NeurIPS Code of Ethics, workers involved in data collection, curation, or other labor should be paid at least the minimum wage in the country of the data collector.

15. Institutional review board (IRB) approvals or equivalent for research with human subjects

Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

Answer: [N/A]

Justification: The research includes no intervention or data collection involving human subjects, so IRB review is not applicable.

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Depending on the country in which research is conducted, IRB approval (or equivalent) may be required for any human subjects research. If you obtained IRB approval, you should clearly state this in the paper.
- We recognize that the procedures for this may vary significantly between institutions and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the guidelines for their institution.
- For initial submissions, do not include any information that would break anonymity (if applicable), such as the institution conducting the review.

16. Declaration of LLM usage

Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does *not* impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

Answer: [Yes]

Justification: LLM-based coding assistants were used for implementation planning, code drafting, adversarial review, regression tests, and copyediting. They are not an evaluated model or data source. The assistance appendix discloses this use; the author froze the protocol before retrieval, reviewed the artifacts, and remains responsible for every claim.

Guidelines:

- The answer [N/A] means that the core method development in this research does not involve LLMs as any important, original, or non-standard components.
- Please refer to our LLM policy in the NeurIPS handbook for what should or should not be described.